Term 2 Checklist: All Topics

Topic	Learning Outcomes	RAG Yourself	
		1	2
Cells	Draw a labelled diagram of an animal cell including: membrane, nucleus and cytoplasm		
	Compare plant and animal cells using a diagram		
	Describe the role cell organelles e.g. membrane, nucleus, cytoplasm, mitochondria		
	Label a diagram to show the parts of a microscope		
	Define specialised cell		
	Identify specialised cells of plants and animals from a diagram		
	Explain how the adaptations of specialised cells are linked to their function		
	Order the levels of organisation in organisms		
	Describe how cells make tissues and tissues make organs		
	State the function and location of key human organs		
	Give three reasons why cell division takes place		
	Sequence the stages of cell division		
	Compare animal and bacterial cells		
	Describe a method to test the levels of bacteria on different surfaces		
	Explain why toothpaste contains anti-bacterial chemicals		
	Define communicable disease		
	State the types of organisms which cause disease		
	Suggest ways we can prevent the spread of communicable diseases		
	Give examples of ways the human body can prevent infection		
	Explain how vaccines can be used to reduce the spread of some diseases		
	Describe the importance of cellular respiration		
	Give the word equation for aerobic respiration		
	Explain why some specialised cells have more mitochondria than others		
	Explain why a yeast solution "bubbles" when sugar is added to it		
	Describe the role of muscles and bones		
	Explain how your arm moves in terms of antagonistic pairs of muscles		
	Describe the effects of exercise on heart and breathing rate		
	Explain why heart rate must increase during intense exercise		
Energy	Name the different forms of energy store e.g. kinetic, thermal		
	Describe the difference between energy stores and transfers		
	State the four ways energy can be transferred		
	Draw energy transfer diagrams for a candle, a catapult and a TV		
	Identify useful and wasted energy in an energy transfer		
	Calculate the percentage efficiency of an energy transfer		
	Describe the difference in terms of energy transfer of an efficient device and a less efficient device		
	State the meaning of "non-renewable"		
	State the tree types of fossil fuel and describe their uses		
	Explain how fossil fuels are used to generate electricity in a power station		
	Define what a renewable energy resource is		
	State examples of renewable energy resources e.g. solar, wind		
	Explain how solar, wind and tidal resources work in terms of energy transfer		
	Compare the advantages/disadvantages of renewables and fossil fuels		
	Draw a diagram showing the carbon cycle		
	Describe how human activities have effected the carbon cycle		

	Describe how carbon dioxide causes a greenhouse effect		
	Explain how human actions have contributed towards global warming		
	List some impacts of global warming		
	Name the three most common gases in our atmosphere		
	State the approximate percentage of each gas in our atmosphere		
	Describe two processes that have caused the atmosphere to change over time		
	Draw a diagram to show how to measure the energy in food		
	Describe the term "sustainable food production"		
	Describe advantages and disadvantages of a vegan diet		
	Describe the difference between organic and intensive farming		
	Describe advantages and disadvantages of organic farming		